

Fill out your details, read the following then do the Bornat Test overleaf

Name (caps) _____ Student ID (9 digits)

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Lab Time _____ Desk Number: Desk- _____

How to work out what a code fragment does: draw a table

The simplest and most reliable way to work out what a code fragment does (and to understand) is to draw a table. This is called tracing or dry running code, Let's work out what this fragment does:

```
int x;           // L1
int y;           // L2
x = 5;           // L3
y = 7;           // L4
x = 10;          // L5
y = x;           // L6
System.out.println(x + " " + y); // L7
```

• Each declaration creates a column for the variable declared.

- So for example to dry run the first part

```
int x;           // L1
int y;           // L2
```

The first line creates a column for line numbers, one for x and the third a column for y. As no value is stored we do not need to add a row. We get the table to the right.

LINE	x	y

• Each assignment places a value in the appropriate column

- So continuing to dry run the next part where we left off

```
x = 5; // L3
y = 7; // L4
```

We put a value 5 in to the column for x on a line with line number L3 Then executing the next instruction, we put a 7 in y's column (line L4)

LINE	x	y
L3	5	
L4		7

• We cross out any old value in a column when that variable is assigned to.

- So continuing to dry run the next part where we left off

```
x = 10; // L5
```

We put 10 in the x column (adding L5 for the line) (This has the effect of crossing out the 5 that was there though crossing it out is not required.) The value you read for a variable is always the LAST value in the column.

LINE	x	y
L3	5	
L4		7
L5	10	

• If we access the value of a variable we take the last value put there (i.e. ignore values crossed out)

- So if the above program continues

```
y = x; // L6
```

We go to the x column and find the LAST value placed there which is a 10. That is its current value. We place the 10 in the y column, in effect replacing the 7 that was already there.

LINE	x	y
L3	5	
L4		7
L5	10	
L6		10

• When something is printed we note it in a n extra print column

- So if the above program continues

```
System.out.println(x + " " + y); // L7
```

The values in x and y are then printed, so the fragment as a whole prints:

10 10

For an actual dry run a single table would be created and added to.

LINE	x	y	printed
L3	5		
L4		7	
L5	10		
L6		10	
L7			10 10

Bornat Diagnostic test: Show all your working

Based on a test created by Saeed Dehnadi and Richard Bornat, Middlesex University

This answer sheet should be saved as a pdf and uploaded on QM+ as your submission for the Bornat Test.

The results do not count for anything. This test is to allow the ECS401 team to diagnose if you need help to ensure they do not fall behind in learning to program. Understanding how assignment works is critical so if you have any problems we want to sort them out now.

Show all your working in the following. Fill out the rows in the tables to justify your answers! No correct table... no marks.

1. What are the new values of a and b after the following code fragment has executed?

```
int  a = 10;  // L1
int  b = 20;  // L2
a = b;        // L3
```

LINE	a	b

a is now _____ **b is now** _____

2. What are the new values of a and b after the following code fragment has executed?

```
int  a = 10;  // L1
int  b = 20;  // L2
b = a;        // L3
```

LINE	a	b

a is now _____ **b is now** _____

3. What are the new values of big and small after the following code fragment has executed?

```
int  big  = 10;  // L1
int  small = 20; // L2
big = small;    // L3
```

LINE	big	small

big is now _____ **small is now** _____

4. What are the new values of a and b after the following code fragment has executed?

```
int  a = 10;  // L1
int  b = 20;  // L2
a = b;        // L3
b = a;        // L4
```

LINE	a	b

a is now _____ **b is now** _____

5. What are the new values of a and b after the following code fragment has executed?

```
int  a = 10;  // L1
int  b = 20;  // L2
b = a;        // L3
a = b;        // L4
```

LINE	a	b

a is now _____ **b is now** _____

6. What are the new values of a, b and c after the following code fragment has executed?

```
int  a = 10;    // L1
int  b = 20;    // L2
int  c = 30;    // L3
```

```
a = b;          // L4
b = c;          // L5
```

a is now _____ **b is now** _____ **c is now** _____

LINE	a	b	c

7. What are the new values of a, b and c after the following code fragment has executed?

```
int  a = 5;     // L1
int  b = 3;     // L2
int  c = 7;     // L3
```

```
a  = c;         // L4
b  = a;         // L5
c  = b;         // L6
```

a is now _____ **b is now** _____ **c is now** _____

LINE	a	b	c

8. What are the new values of a, b and c after the following code fragment has executed?

```
int  a = 5;     // L1
int  b = 3;     // L2
int  c = 7;     // L3
```

```
c  = b;         // L4
b  = a;         // L5
a  = c;         // L6
```

a is now _____ **b is now** _____ **c is now** _____

LINE	a	b	c

9. What are the new values of a, b and c after the following code fragment has executed?

```
int  a = 5;     // L1
int  b = 3;     // L2
int  c = 7;     // L3
```

```
c  = b;         // L4
a  = c;         // L5
b  = a;         // L6
```

a is now _____ **b is now** _____ **c is now** _____

LINE	a	b	c

10. What are the new values of a, b and c after the following code fragment has executed?

```
int  a = 5;    // L1
int  b = 3;    // L2
int  c = 7;    // L3

b  = a;        // L4
c  = b;        // L5
a  = c;        // L6
```

a is now _____ **b is now** _____ **c is now** _____

LINE	a	b	c

11. What are the new values of a, b and c after the following code fragment has executed?

```
int  a = 5;    // L1
int  b = 3;    // L2
int  c = 7;    // L3

b  = a;        // L4
a  = c;        // L5
c  = b;        // L6
```

a is now _____ **b is now** _____ **c is now** _____

LINE	a	b	c

12. What are the new values of a, b and c after the following code fragment has executed?

```
int  a = 5;    // L1
int  b = 3;    // L2
int  c = 7;    // L3

a  = c;        // L4
c  = b;        // L5
b  = a;        // L6
```

a is now _____ **b is now** _____ **c is now** _____

LINE	a	b	c